NET SOFTWARE

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Introduction

This software was the brainstorm of Bjarne Bäckström. His idea was to allow a .dsk file on a PC to look like a disk drive to FLEX. The PC can open .dsk files and make them available to the FLEX system as another disk drive. The communications are by means of a serial link. Bjarne worked on the 6809 end, and Ron Anderson wrote the PC end of the system. Joel Setton requested some additions to the PC end and wrote the Flex end for additional utilities. Joel and Ron worked to test and bullet-proof the software to make it feasible to run on a minimal PC with no monitor or keyboard. Mickey Ferguson acted as the Beta-tester, identifying a few bugs and providing suggestions for additional improvements.

This document explains the benefits in using Netpc, the way to adapt it to your system, and contains a detailed manual of all the utilities provided with Netpc. We trust you will enjoy this set of programs.

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1 - WHAT NETPC CAN DO FOR YOU

The core idea of this toolset is to make a DSK file, stored on a PC, look like a drive to a Flex system. When NETPC is installed on your PC and your Flex system, these being connected through a serial link, you can select any DSK file on the PC and "mount" it to make it look as a Flex Drive. You can ASN W=X and use the .dsk file as a working disk. You can now copy files from this "virtual" drive to drive 0 or 1 just as though the PC were a real disk drive. You can run BASIC and load the program from it, etc. In short, anything you can do in FLEX that requires disk access can be done on the NetPC drive as well as on any of the physical drives on the Flex computer.

Think of how an old, obsolete PC can extend the capabilities of your Flex system:

- You can use the PC to store your Flex working files. For example, a floppy with 35 tracks and 10 sectors per track, will be stored as an 89600 byte DSK file on the PC. NetPC will read and write to the DSK file just as it would read/write on a regular floppy.

- You can also use the PC as a giant library of FLEX software; for example, it could contain DSK images of your favorite FLEX products. Any of these "virtual floppies" can be mounted and available as a virtual Flex drive with just a few keystrokes, which is much easier and quicker than taking a regular floppy out of its box and inserting it into the drive!

- Once a DSK file is stored on the PC, you can easily E-mail it (or the other way around if you receive a DSK file)

- You can backup that collection of old floppies on the PC, and later re-create a floppy from its DSK image. The re-created floppy will be a perfect clone of the original one, including boot sector, SIR, random files, free sector chain and everything.

- You can also backup the whole set of DSK files to a writeable CD if the PC has a CD-RW drive.

- In short, you can use the PC as a giant store of Flex disks. Even the largest possible drive in Flex (256 tracks of 256 sectors, a total of 16 megabytes) is small with regard to the amount of storage space available on a PC. Or, storing a hundred Flex floppies only needs 20 to 40 megabytes!

You can even run NetPC on a very reduced PC configuration, without keyboard, mouse or monitor, to act as a "Flex file server". The way to do this is described in Section 3.

In most of this manual, drive #3 will be used as an example; however, any of the Flex drive numbers (0 to 3) can be used as the Netpc virtual drive.

2 - ADAPTING TO YOUR SYSTEM AND FIRST BOOT

The software adaptation involves editing the driver which will run on your Flex system, to make it compatible with the serial port which will be connected to the PC; once assembled, this driver will be specific to your Flex system. On the PC side, the adaptation is simpler to do since the interface is a standard COMM port.
Once NetPC is running for the first time, you can easily transfer the utility files which are provided, and which will not need any adaptation.

2.1. Flex hardware: Your Flex system needs to have one free serial port, if possible using a 6850 ACIA chip, and including a baud rate generator to run the ACIA at 9600 bps or higher; obviously, the higher the baud rate, the better. The system has been tested at 57600 baud, at which speed the "virtual disk" addressed through Netpc seems almost as fast as a local hard disk. The serial link to the PC needs to include only three wires: Tx, Rx, and GND. The /CTS pin of the ACIA (pin 24) must be driven LOW, either by a direct connexion to ground, or by the incoming CTS line, in which case the serial cable needs one additional wire for CTS.

If the serial interface chip is any other than a 68(X)50, the source file NETDRV.TXT must be adapted to match the particular chip which is used.

2.2. FLEX software:
2.2.1. You first have to copy two text files, NETDRV.TXT and ACIA.TXT to your Flex system. This can be done through a serial link, or with a floppy using the MS-DOS utility which is available on the FUG website. (or you could also retype them in Flex without comments and later transfer the full files with comments after the system is running.)

2.2.2 Then you have to edit one file, ACIA.TXT.
* Edit the register addresses to match those of your ACIA;
* Also set the programming of the baud rate generator to supply the ACIA with the proper clock in 16x mode. For example, if you want to run the serial link to the PC at 9600 baud, the baud rate generator must supply a 153.6 kHz clock. A standard 6850 ACIA can run up to 38400 bps in 16x mode, the 68A50 and 68B50 can both run up to 57600 bps.

If your serial port uses a fixed or "strapped" baud rate generator which is not programmable, you may remove the two lines of code in ACIA.TXT which program the baud rate generator. NetPC uses a fixed baud rate and does not need to change the baud rate at any time.

2.2.3. Once these changes are done, assemble NETDRV.TXT to NETDRV.CMD on your Flex system drive. The assembly will call ACIA.TXT as a library file, so NETDRV.CMD will be able to use the baud rate generator and ACIA at the appropriate addresses for your system.

2.2.4. Alternatively, the above tasks (2.2.1 to 2.2.3) can be done on one of the Flex emulators running on your PC. The same things must be done: edit ACIA.TXT and assemble NETDRV to generate NETDRV.CMD, which must then be copied to your Flex system drive.

2.3 PC software
2.3.1. Locate the following five files among those in the unzipped package:
- Netpc.exe, which is the executable which will run on the PC
- blank.dsk, which is an empty DSK file
- utils.dsk, which contains several Flex utilities
- sources.dsk, which contains the source code of these Flex utilities
- config.txt, which configures Netpc, as explained in the next paragraph.
Copy these five files to the directory from which you want to run NETPC, e.g. c:\netpc\ on the PC.

2.3.2. On the PC, edit C:\config.txt as follows:
- The first line is the number of the PC's comm port which will be connected to Flex when running Netpc. This can be 1 or 2.
- The second line contains the desired baud rate for the serial link, and must obviously match the one in the ACIA.TXT file (note: there is no 16x division factor on the PC; for 9600 bps, this line should contain 9600)
- The third line is "S" or "F", depending on whether you are using a Slow or Fast PC.
- The fourth and last line contains either "S" for normal use, or "V" for debugging purposes. Set it to "V" for the moment.

2.4. Serial link: Connect the Flex serial port to the appropriate comm port on the PC.

2.5. First boot-up

2.5.1 On the PC: exit Windows and reboot under DOS. NETPC controls the serial port directly and it WON'T run correctly in a DOS window in Windows 95/98. On the PC, go to the C:\ directory, and start netpc by typing:

```
netpc utils
```

2.5.2. On your Flex system, type NETDRV. It will report the following:

```
NETDRV version X.X
Serial connection established
Current MSDOS drive is C:
Remote .DSK drive installed as drive 3
```

At that point, you have successfully started Netpc!!!! The following steps will install all the utility commands which come with Netpc, and get your systems ready for normal use of Netpc after this initial boot.

2.5.3. On your Flex system, type DIR 3. You will see that Drive #3 now contains the 9 command files contained in C:\utils.dsk on the PC. At this point, NETPC is running, so these files can be accessed on drive 3 as if they were on any Flex drive. There are nine utilities in .CMD form: RMOUNT, RESYNC, RDELETE, RCD, RDRIVE, RLIST, RDIR, RCREATE and REXIT.

These nine command files are ready for use and need no customization. Copy them all from drive 3 to your Flex system drive.

At that point, the Netpc toolset is ready, and the use of the utilities is described in detail in the User's manual (section 4). You can configure NetPC to its "silent" mode, by editing the last line of CONFIG.TXT and replacing the "V" with an "S". In this mode, the debugging messages are not displayed by the PC.

3 - NORMAL USE

Once the above adaptations are made, everyday use of Netpc is simple and can be made as automatic as needed.

There are at least two ways of starting Netpc on the PC:

- Manually: under DOS (not a DOS window in W9X), at the c:\ prompt, type "netpc blank"

This will start Netpc and open the file "blank.dsk" which will be mounted as drive #3 for Flex. Netpc always needs a DSK file name as a start parameter. This file will be mounted during the startup phase. Of course, after NETPC is running, another DSK file can be mounted by using the RMOUNT command from Flex.

- Automatically: The PC can boot from a DOS or Win9X bootable floppy, containing an appropriate autoexec.bat file:

```
set comspec=c:\command.com
c:\
netpc blank
```

Here also, Netpc needs a DSK file as a parameter when starting. When this is done, the PC will boot on the diskette, with no user input required through the keyboard or mouse, and no monitor is needed. This mode of operation allows a minimal and very inexpensive PC configuration to be used as an unattended Flex file server.

After Netpc is started on the PC using either method, the user should start NETDRV on the Flex system to get everything up and running. This can be done manually or as part of the STARTUP.TXT file.

If NETDRV is run without any command line parameters, the NetPC drive will be drive #3 by default. However, the user can specify another drive number, for example NETDRV 2 will configure drive #2 as the Netpc drive.

The software should always be launched first on the PC, then on Flex.

RCD Remote Change Directory

This command will change the current directory, which is the default directory which the other utilities will use. For example, if no file path is provided in the RCREATE command, the new file will be created in the current directory.

Syntax: RCD <filepath>

Example: RCD C:\folder3

RCREATE Remote Create

The RCREATE command will prompt the user for the location, name, and other parameters to create a new DSK file. After all parameters are entered, the user will be asked for a confirmation (yes/no) and if the answer is positive the file will be created.
No parameters are required in the command line.

Example:

```
+++RCREATE
File path ? C:\folder3
File name ? newname
Disk number (in decimal) ? 123
Number of tracks (in decimal) ? 35
Sectors per track (in decimal) ? 10
OK to proceed ? Y

File successfully created
+++
```

NOTES:
- The file path is the directory where the new file will be created. If none is specified, the file will be created in the current directory.
- The file name ("newname") will be used for both the name of the DSK file, as known by MS-DOS, and for the volume name which is stored in the SIR (track 0, sector 3) of the DSK file.
- The disk number, 123 in this example, is stored as the volume number in the SIR.
- If non-numeric values are entered for the disk number, the number of tracks or the number of sectors per track, the question will be repeated.
- RCREATE does not check for the "validity" of the track/sector format entered by the user. The file will be created, as long as the values given are numeric.
- In the above example, the disk image will be created with 35 tracks (including track 0) of 10 sectors, for a total of 340 data sectors.
- Note that RCREATing a disk does not automatically RMOUNT it or change the current directory. You must use RCD to change directories if needed, and RMOUNT to mount the new disk.

RDELETE Remote Delete

RDELETE will prompt the user for the file path and the name of the file to be deleted. If no file path is given, the current directory will be assumed. RDELETE also adds a .DSK extension to the filename.

RDELETE will then display the full path and name of the file, and will prompt the user for a response (Yes/No) before deleting the file.

If the file cannot be found, an error message will be displayed.

Syntax: RDELETE <filename> [cr]

RDIR Remote Directory

RDIR performs a DIR command under DOS, and returns an exact copy of the listing generated on the MS-DOS screen:
- The current directory is used
- Only files with the extension .DSK will be listed.
- Command line parameters are allowed, and are passed to the DOS command.
- RDIR displays 20 lines at a time, then prompts the user before displaying the next 20 lines. Typing a space will display the next lines, typing [esc] will stop the listing and return to the Flex prompt.

Examples

+++RDIR [cr]	Lists all DSK files in the current directory
+++RDIR A2* [cr]	Lists all DSK files which start with A2.

RDRIVE Change drive

RDRIVE instructs the MS-DOS program to point to another drive, thus offering the ability to access any DSK file on any drive.

Syntax: RDRIVE [letter]

If no drive letter is specified, RDRIVE will display the current drive and directory path.

RESYNC Re-SYNChronize

If something should happen and you restart the PC, it is looking for the Sync information from the FLEX system. RESYNC will re-establish the communication link and you can proceed. The command is simply: RESYNC. If all is working properly you should not have to do anything at the PC end of the link.

Syntax: RESYNC [cr]

RLIST Remote List

RLIST display a list of all the directories which are contained in the current directory. It is provided as an aid to the user when "navigating" in the MS-DOS directory tree structure. No parameters are needed. RLIST displays the listing in the same fashion as RDIR, i.e. 20 lines at a time, then prompting the user for more.

Syntax: RLIST [cr]

RMOUNT Remote Mount

RMOUNT will "mount" a DSK file to drive 3, in other words it selects a DSK file, opens it and sets it as Drive #3. After a file has been RMOUNTed, it can be read or written to by Flex as drive #3 as if it were a standard floppy drive. The user may specify just the name of the file, in which case RMOUNT assumes the file is in the current directory. Otherwise, the user may type the complete path and file name to RMOUNT a file which is in another directory.
RMOUNT does not require a file extension to be typed; however, if one is entered, it must be .DSK (lowecase letters OK) or an error message will be generated.

NOTE: There is no need to "unmount" a file before RMOUNTing another one. However, a file which is "mounted" is open under MS-DOS, and may be corrupted if the PC is powered down while the file is open.

Examples:

+++RMOUNT <filename.dsk> [cr]

+++RMOUNT <filename> [cr]
Mount a file which is in the current directory. NETPC will automatically add the .DSK extension to the file name.

+++RMOUNT c:\folder\filename [cr]
Mount a file which is in any directory. In this case the file will be mounted without changing the current directory.

REXIT Remote Exit

This command will close all files open on the PC and exit to DOS. On the Flex system, it will uninstall the NETDRV drivers: restore the Flex drivers to their original values (i.e. what they were before NETDRV was run), clear the memory space used by the NETPC drivers, and restore MEMEND to make the memory space available again for other programs.

Note: MEMEND will be changed ONLY if no other program has changed it after NETDRV was loaded; in other words, memory will be freed only if the NETDRV code is right "above" MEMEND; see section 6.3 for more detail.

5 - INTERNAL OPERATION

When the NETPC program is started on the PC, it initializes ports and waits for a character on the serial line. It echoes the character received, and waits for another which it also echoes. When it receives an $AA it echoes that and drops out of the character loop ready to run. At that point in verbose mode it prints "Communications Established" on the PC screen, reads the SIR of the .dsk file mounted as default, and waits for a command character from the FLEX end.

NETDRV: This command file is run on the Flex system, and contains two parts:
- The "drivers" part which is initially loaded in low memory, and contains the serial in/out routines to communicate with the PC, plus the disk drivers for the virtual drive which is on the PC and accessible to Flex as drive #3; this part of the code must be position-independent.
- The "installer" part which runs when NETDRV is started, and checks for a working serial communication with NetPC (which must then be running on the PC).

If the communication is found to be OK, NETDRV will relocate the "drivers" part to the top of memory where it will stay resident; MEMEND will be decremented accordingly. Thus, the disk drivers and serial drivers will remain available between the new MEMEND and the end of physical memory (which is $BFFF in most systems). The size of these drivers is close to 512 bytes, thus in a system which has 48K of memory, MEMEND gets decremented down to approximately $be00 . These drivers are used by the utilities supplied with NetPC, to communicate with the PC.

The "installer" part loads in the Flex command area ($c100) and is useful only for launching NETDRV and initializing the serial communication with the PC. After this is done, it can be overwritten.

Signals between Flex and PC:
Each utility sends a different command character from the Flex end of the combination. A description of the command and what is sent in each direction is given below:

RESYNC The flex end sends $55 and then $AA and expects to have them echoed. If they are, it assumes a connection is made.

RCD sends "P" followed by the desired path specified in the command line and a <CR>.
The PC then sends ACK if it successfully changed the directory and NACK if not.

RDIR sends "A"
PC gets a directory listing of all *.dsk files if there is no parameter specified in the command line.
If there is a parameter in the command line (with or without PC wildcard characters) the PC returns matching filenames.
The PC sends one line of directory information at a time and waits for a character from the FLEX end: ESC terminates the process, any other character gets another line of the directory until all lines are sent. Then ACK is sent by the PC.

RLIST sends "I"
PC gets a list of the subdirectories of the current directory. As in RDIR, one line at a time is sent. ESC from the flex end terminates the process. Any other character gets another line until all the lines are sent. Then ACK is sent.

RDELETE Flex sends "D" followed by the command line parameter, the filename of a file in the current directory. The PC tests to see if the file exists and returns NAK if it does not. If it does exist it is deleted and ACK returned.

RCREATE sends "C" and all of the information prompted for by RCREATE:

```
Path (empty = current directory) <CR>
Filename<CR>            (filename = disk name)
Disk Number<CR>         (in ASCII, 1 to 4 digits)
Number of tracks<CR>    (in ASCII)
Sectors per Track<CR>   (in ASCII)
```

the PC then creates a .dsk file of the name Filename and of the specified size. It returns ACK unless for some reason it can't open the output file. It returns NAK in that case.
A CREATEd disk must be MOUNTed before it can be used.

RMOUNT sends "M" followed by filename<CR>
PC closes the presently mounted .DSK file and opens the new one. If the disk file is not found, or is write protected, NAK is returned. If the file is mounted successfully, ACK is returned.

ALL PARAMETERS FOLLOW THE COMMAND LETTER WITHOUT A SPACE

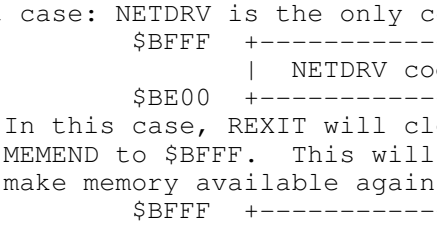
6 - MISCELLANEOUS NOTES, THOUGHTS AND CAVEATS

6.1. When the selected MSDOS drive is a floppy (A or B), Netpc will use longer time-outs to match the slow speed of a floppy, as compared to a hard disk. Nevertheless, when performing a very long task such as RCREATING a 1000+ sector DSK file on a floppy, the timers imbedded in the Flex drivers may time-out. If this happens, typing a RESYNC command in Flex after the file creation is complete on the PC will get everything back on track.

6.2. When NETDRV <drive number> is launched on the Flex system, it defaults to drive #3 if no drive number is specified, or if a number is specified the drive with that number will be the NETPC drive. After NETDRV starts, there is no way to change the drive number. The only way to change to a different drive number is to use REXIT or to reboot Flex, then start NETDRV with the desired new drive number.

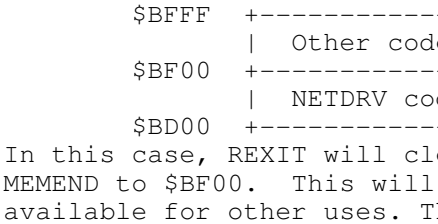
6.3. Some additional notes on how REXIT frees memory: three cases can occur.

6.3.1 First case: NETDRV is the only code in high memory.

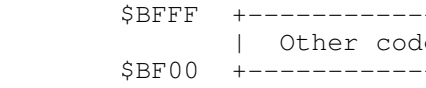


In this case, REXIT will clear the NETDRV code area, and will set MEMEND to $BFFF. This will effectively increase MEMEND and will make memory available again for other uses.

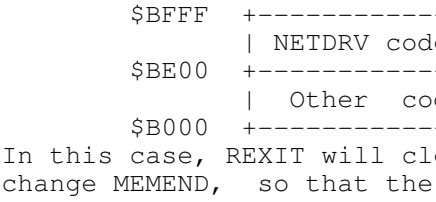
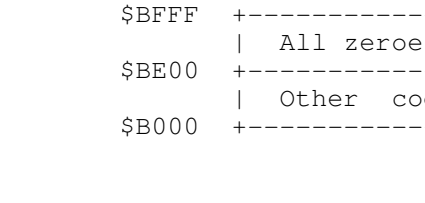
6.3.2 Second case: NETDRV was loaded after some other code in high memory.



In this case, REXIT will clear the NETDRV code area, and will set MEMEND to $BF00. This will increase MEMEND and will make memory available for other uses. The "other code" will not be affected or moved in any way:



6.3.3 Third case: Some other code was loaded in high memory after NETDRV.



In this case, REXIT will clear the NETDRV code area, but will not change MEMEND, so that the "other code" will not be moved. The "other code" is not assumed to be position-independent. This implementation was chosen so that the functionality of any other software loaded in this area will not be impacted.